Data Report

Narrative Homogenization in Hollywood Films

Team 3
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1. Introduction

This report documents the data acquisition, preprocessing, and exploratory data analysis (EDA) conducted for the Narrative Homogenization Pipeline. The goal is to determine whether Hollywood films have become narratively more similar over time by analyzing film scripts across four cinematic eras spanning 1911 to 2026.

All scripts have been preprocessed and deduplicated by the team prior to this analysis. The pipeline reads directly from cleaned .txt files and produces quantitative features used for both unsupervised clustering (BERT + KMeans) and supervised classification (Random Forest).

2. Data Source

2.1 Origin

Film scripts were sourced from two pipelines:

Pipeline 1 — ScriptHive (PDF):

Scripts were scraped from ScriptHive via reverse-engineered Supabase API calls. A custom signed-URL downloader (scraper.py) was built to authenticate and download PDFs. 12,322 PDFs totaling 32GB were downloaded.

Pipeline 2 — Pre-processed TXT (current):

A curated dataset of cleaned .txt script files was assembled by the team, deduplicated, and stored in Google Drive. This is the primary dataset used for all modeling.

```
Path: processed_scripts/cleaned_scripts/sorted_by_duplicates_2/unique/
```

2.2 Raw Dataset Summary

Property	Value
Total raw PDF files scanned	12,322
Total PDF dataset size	32.0 GB
Total curated TXT files	4,225
Files with parseable year	2,164
Files excluded (no year found)	2,061
Final usable scripts	2,056

2.3 Filename Structure

Scripts follow a semi-structured naming convention:

```
Title_Writer(s)_Type_MM DD YYYY___Draft___.txt
Example: Strange Days_James Cameron_Film_8 11 1993_____.txt
```

Year is extracted via regex and capped between 1920 and 2026 to exclude false matches.

3. Era Definition

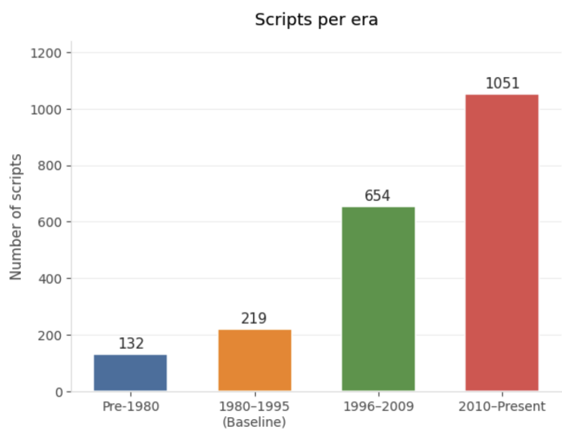
Films are assigned to one of four cinematic eras based on release year:

Era Label	Year Range	Role in Analysis
Pre-1980	Before 1980	Historical reference
1980-1995 (Baseline)	1980-1995	Baseline for homogenization comparison
1996-2009	1996-2009	Mid-era
2010-Present	2010-2026	Modern era — primary comparison target

4. Exploratory Data Analysis

4.1 Scripts per Era and Era Share

The charts below show the count and proportional share of scripts across the four cinematic eras. The 2010-Present era dominates at 51.1%, while the 1980-1995 baseline era accounts for only 10.7% of all scripts — creating the class imbalance addressed in the modeling phase via SMOTE oversampling.



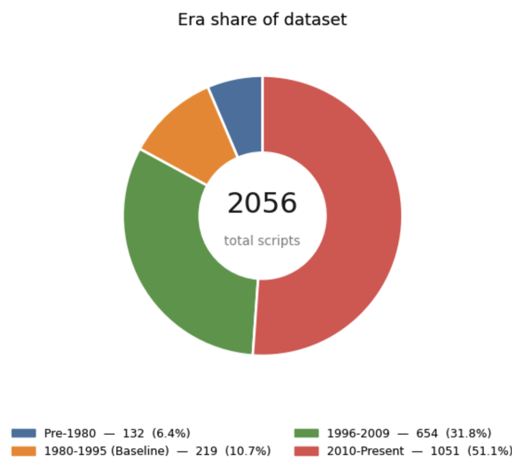


Figure 1: Scripts per era (bar chart) and proportional era share (donut). The baseline era 1980-1995 accounts for only 10.7% of the dataset.

Era	Scripts	Share
Pre-1980	132	6.4%
1980-1995 (Baseline)	219	10.7%
1996-2009	654	31.8%
2010-Present	1,051	51.1%
Total	2,056	100%

4.2 Scripts per Decade and Class Distribution

The decade-level breakdown reveals the dataset is concentrated in the 2000s (525 scripts) and 2010s (847 scripts). The class distribution panel confirms a 4.8:1 imbalance between Modern and Baseline eras.

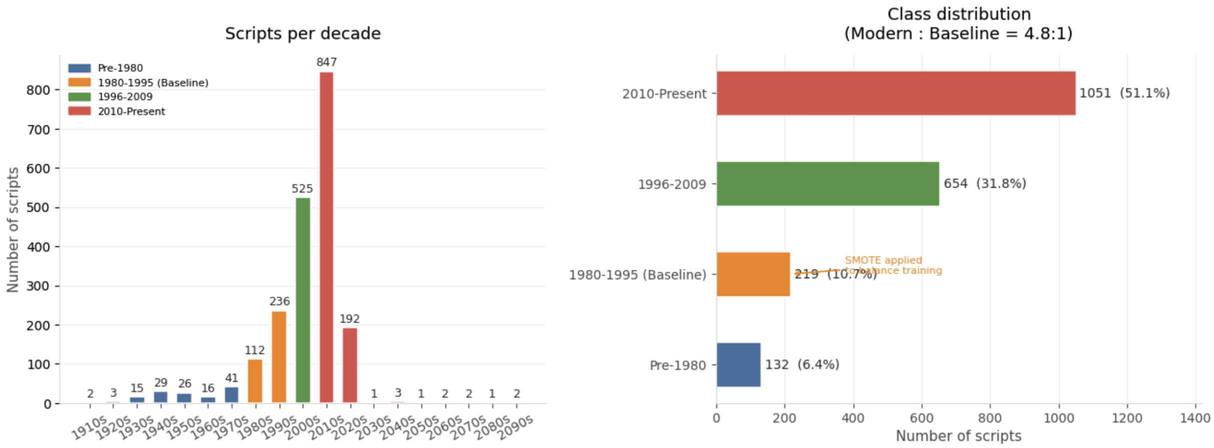


Figure 2: Scripts per decade (left) and horizontal class distribution showing 4.8:1 imbalance (right). SMOTE is applied to balance the training set.

4.3 Release Year Distribution

The histogram below shows script count by release year in 2-year bins. The distribution peaks sharply around 2008-2013 and drops off in recent years due to publication lag. Pre-1980 scripts are sparse, limiting historical comparison depth.

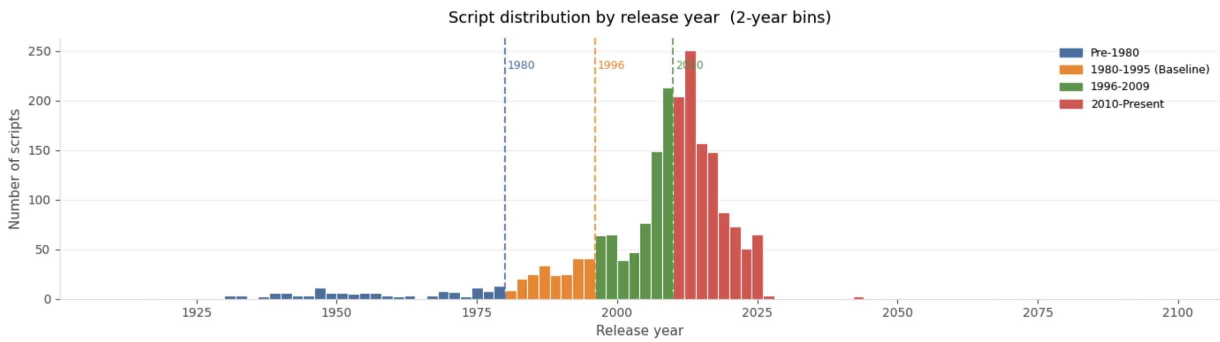


Figure 3: Script count by release year in 2-year bins colored by era. Dashed lines mark era boundaries at 1980, 1996, and 2010. Distribution peaks around 2008-2013.

- Very few pre-1980 scripts are available — limits historical comparison depth
- 2009-2013 is the most densely represented period in the dataset
- Recent years (2020-2026) drop off due to publication lag

4.4 Word Count Distribution by Era

A minimum threshold of 2,000 words was applied during loading. The overall mean word count is 23,118 words. Distributions are broadly similar across eras, suggesting consistent screenplay length norms — an initial signal that surface-level script length has not changed significantly over time.

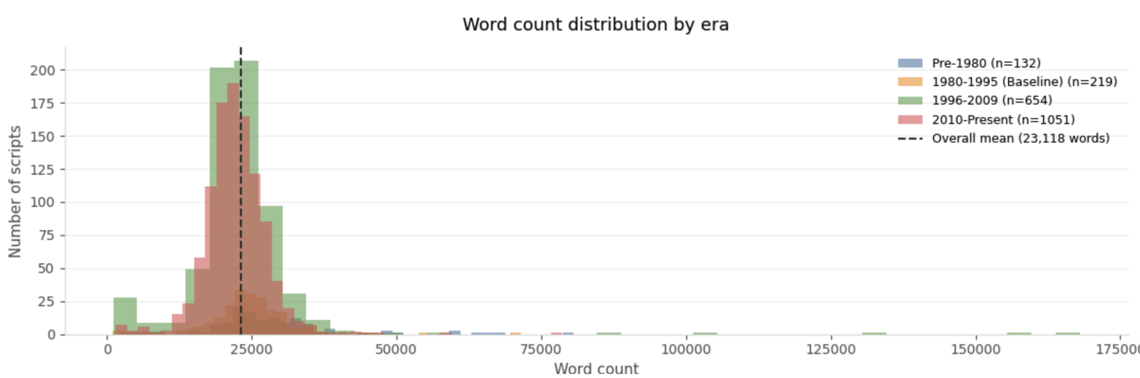


Figure 4: Word count distribution by era. Overlapping distributions centered around 20,000-25,000 words. Overall mean = 23,118 words (dashed line).

Metric	Value
Minimum word count threshold	2,000 words

Overall mean word count	23,118 words
Peak of distribution	~20,000-25,000 words
Right-tail outliers	>75,000 words (rare, likely transcripts)

5. Preprocessing

Raw script text undergoes the following cleaning steps before any feature extraction or modeling. The goal is to retain only dialogue and action lines and remove all screenplay formatting noise.

Step	What is removed	Reason
Scene headings	INT. COFFEE SHOP - DAY	Formatting noise
Character cues	ALL CAPS lines under 40 chars	Speaker labels, not dialogue
Parentheticals	(beat), (quietly)	Stage directions
Page numbers	Lone digits on a line	PDF artifact
Transition directives	FADE IN, CUT TO, SMASH CUT	Formatting noise
Copyright lines	Lines with copyright, WGA, registered	Legal boilerplate
Separator lines	--- and === sequences	Visual dividers

On average, preprocessing removes 15-20% of raw token count, retaining only narrative-bearing text. This improves the quality of downstream VADER sentiment scoring and BERT semantic embeddings.

6. Data Limitations

Limitation	Description	Mitigation
Class imbalance	Baseline era has 4.8x fewer scripts than modern	SMOTE oversampling in training
Year parsing gaps	49% of TXT files have no parseable year	IMDb metadata lookup planned
Filename inconsistency	Loose naming convention causes misparses	Year capped 1920-2026
Geographic bias	Skews toward English-language Hollywood	Acknowledged as scope limitation
Recency bias	2020-2026 underrepresented due to pub lag	Noted in findings interpretation

7. Files and Locations

File	GitHub Location	Description
cell3_txt_loader.py	Code/Data_Acquisition_and_Understanding/	Loads TXT scripts into df
data_dashboard.py	Code/Data_Acquisition_and_Understanding/	Generates EDA visualizations

data_dashboard.png	pipeline_outputs/	Main 4-panel dashboard
year_distribution.png	pipeline_outputs/	Year histogram by era
wordcount_distribution.png	pipeline_outputs/	Word count distribution
film_features.csv	pipeline_outputs/	Extracted features per script

8. Next Steps

- Augment pre-1980 and 1980-1995 scripts to reduce class imbalance
- Investigate IMDb metadata API to recover years for the 2,061 unmatched files
- Add genre labels from IMDb to enable genre-stratified homogenization analysis
- Run full feature engineering pipeline on the 2,056-script dataset
- Add Jensen-Shannon Divergence as a vocabulary convergence metric
- Add rare-word sentiment arc as an improved feature over standard VADER arc